

## Week 3 Report: Literature Review and Dataset Collection (final week), Data Preprocessing (start)

Project: Determining the Role of Big Data Applications in Fraud Detection and Financial Security in Financial Institutions of Ireland

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### What I did this week

This week I finished the literature review and dataset collection and started preprocessing. For the review I added Ma et al. (2026), who warn that resampling should be judged on a test set with the real class balance, and Horobets et al. (2025) on EU rules for AI in banking.

For preprocessing, my first step was the duplicates I found in Week 2. The file has 284,807 rows and 1,081 of them are exact copies. I checked what happens if they are left in: after a normal 70/30 split, 456 test rows also appear in the training data, 7 of them fraud. So I removed duplicates before splitting. The final split is 198,608 training rows (331 frauds) and 85,118 test rows (142 frauds), stratified with seed 42. I added a check to the code that stops the run if any test row is also in training.

| Step                                | Result        |
|-------------------------------------|---------------|
| Rows in file                        | 284,807       |
| Duplicates removed                  | 1,081         |
| Rows after cleaning                 | 283,726       |
| Training rows (frauds)              | 198,608 (331) |
| Test rows (frauds)                  | 85,118 (142)  |
| Test rows leaked if duplicates kept | 456           |

### Next week

Scale Time and Amount, add an hour of day feature and explore the cleaned data.